Name:_____

MZUZU DIOCESE

2023 MSCE MOCK EXAMINATIONS

BIOLOGY

Thursday, 16 March

Subject Number: M012/II
Time Allowed: 1½ hours
(8.00am onwards)

PAPER II

(100 Marks)

(Practical)

Instructions:

1. This paper contains **7 pages**. Please check
2. Answer all questions in the spaces provided.
3. Write your *name* and *school* on top of every page.
4. In the table provided on this page, tick against the question number you have answered.
5. Hand in your paper to the invigilator at the end.

Question Number	Tick if Answered	Do not write in these margins	
1			
2			
3			
4			

Name:_____

Section A (20 marks)

Answer all questions in this section

1. Describe an experiment you would carry out to measure the uptake of water by a plant by means of photometer. In an essay form your answer should include **procedure, results** and **conclusion**

Name:_____

_____

_____(10 marks)

2. **Figure below** is a method used to estimate plant population in an ecosystem.

Explain how the method is used to estimate plant population. In an essay form your answer should include procedure and conclusion.

_____

_____

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_____(10 marks)

Name:_____

Section B (20 Marks)

Answer all questions in this section

3. You are provided with the following materials

- A grass stem with leaves labelled A
- Bidens pilosa (chisoso) stem with leaves labelled B
- A razor blade or knife
- A hand lenses

Cut a cross section of specimen A and B

a. (i). Draw the cross section of the specimen A and B showing the arrangement of vascular bundles

(4marks)

(ii) Describe the arrangement of vascular bundles in the drawings of the specimen above

_____

_____ _(2marks)

Name:_____

b. To which groups of flowering plants of flowering plants do the two belong?

A_____(1mark)

B_____(1mark)

c. Give any two structural differences between leaves of specimen A and B

_____

_____

_____

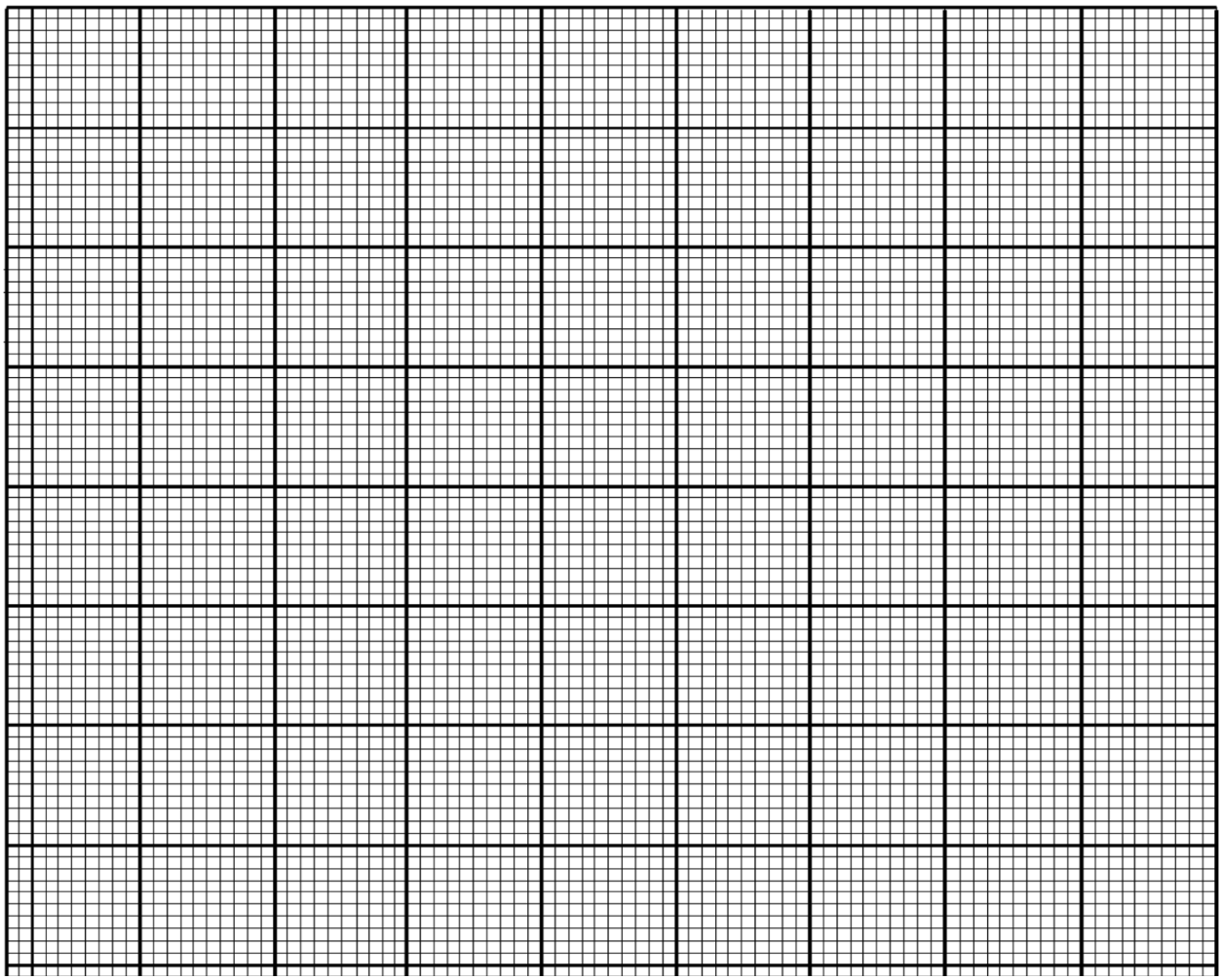
_____(2marks)

4. Table 1 shows results of an investigation on the effect of light intensity on the rate of photosynthesis in a water plant. The rate of photosynthesis is measured by the number of air bubbles produced per minute. Use it to answer questions that follow.

Distance of a lamp from plant (cm)	Number of air bubbles per minute
0	29
10	28
20	27
30	24
40	20
50	12
60	6

Name:_____

a. plot a graph of number of air bubbles produced per minute against distance of the lamp from the plant.



(4marks)

b. i. What is the relationship between the number of air bubbles produced and the distance of the lamp from the

_____

_____(2marks)

Name:_____

ii. From the graph, find the number of bubbles produced when the distance of the lamp from the plant is 35 cm

_____(1mark)

c. mention the type of gas contained in the air bubbles produced in the investigation

_____(1mark)

d. explain how the gas mentioned in (c) is produced.

_____

_____

_____(2marks)

END OF QUESTION PAPER

NB: This paper contains 7 pages